

Strategic Planning for Healthier and Safer Dams

Accounting for schistosomiasis transmission risk in the development of African water management infrastructure

Dams, development and human health

Water management infrastructure - including dams, reservoirs, and irrigation systems - is essential for enhancing food security and clean energy, particularly in regions of sub-Saharan Africa vulnerable to climate change. However, these projects may inadvertently elevate the risk of transmission of water-associated diseases, disproportionately affecting marginalized communities [1].



This small dam on the Senegal river, under the Kayes' bridge in Mali, is always crowded with people engaged in their daily chores. PHOTO by: Evgeni Zotov (FLICKR)

The Problem: Dams can heighten disease risk

Rural communities in sub-Saharan Africa are exposed to the parasites that cause schistosomiasis, also called Bilharzia or snail fever. These parasites rely on snail vectors that thrive in freshwater environments, including streams, lakes, ponds, irrigation canals, and dams.

It is well established that the development of dams and irrigation systems increases the risk of schistosomiasis transmission by creating suitable freshwater habitats for the snails [2].

Despite a large body of scientific evidence, the risk of schistosomiasis transmission is generally overlooked in Environmental Impact Assessments (EIAs) of dams and irrigation schemes. EIAs are the normative tool adopted by nearly every country in the world to assess the environmental and health consequences of major infrastructure projects.

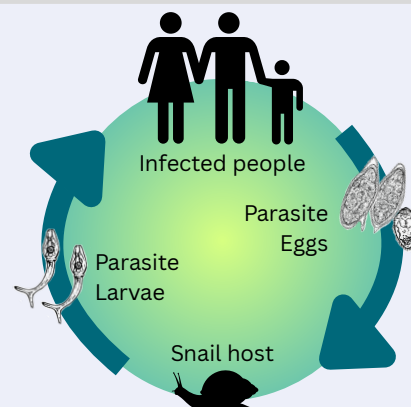
The Solution

Here we propose a pathway for the systematic integration of schistosomiasis risk assessments into EIAs and outline simple and cost-effective engineering, environmental and public health strategies to mitigate the risk of schistosomiasis transmission associated with water infrastructure. Incorporating health impact assessments (HIAs) into water management infrastructure planning is crucial to curb disease transmission while supporting sustainable development goals.

Schistosomiasis: Key facts

Globally, 800 million people at risk and 250 million people are infected. Most infections are in school-age children, requiring medical treatment on an annual basis. Schistosomiasis is second only to malaria and dengue for disease burden among the neglected tropical diseases.

Schistosomiasis causes fever, enlarged lymph nodes, abdominal pain, blood in the urine, & difficulty or pain when urinating. It damages the liver, kidneys, bladder, and spleen.



In areas up to 30 km from dams, the poorest households, which engaged in subsistence agriculture, experience a **32-45%** average loss of income due to schistosomiasis. [4]

With more than 600 medium and large dams in construction or proposed across Africa, alongside many more small reservoirs to support irrigation for agriculture expansion and intensification, these water management infrastructure could affect the health of millions of people.

References: [1] Rohr et al. 2019 *Nature*. [2] Steinmann et al. 2006 *Lancet Infect Dis*. [3] Rinaldo et al. 2021 *Inf. Dis. Pov*. [4] Sow et al. 2002 *Ann. Trop. Med. Par*.

More than **600 dams** are in construction or are being proposed. Addressing disease risk can benefit communities across Africa.

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Diamna Dam, a Cautionary Tale

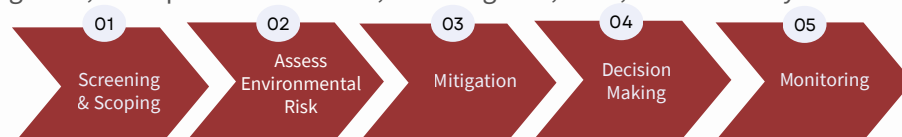
Completed in 1986, the Diamna Dam at the mouth of the Senegal River aimed to combat drought, prevent salt intrusion, and boost agriculture. Before its construction, schistosomiasis was uncommon. Soon after, cases surged, including intestinal schistosomiasis previously unseen in the region. Within two decades, the lower basin became a global hotspot, revealing a clear link between dam development and disease. To prevent such outcomes, disease monitoring and control must be integrated into the planning of water management infrastructure. For example, fish and prawn ladders can be built with the dam so natural predators of the snails continue to persist.

Successful Control Efforts in Egypt

Until the 1930s, urogenital schistosomiasis was very common in Egypt, with peaks of around 60% of people suffering from the disease in specific regions, such as the Nile Delta and the Valley. The country then implemented a highly successful integrated control programs including: preventive chemotherapy treatment in the form of mass drug administration, drainage projects, concrete lining of irrigation ditches, and biological control of snail vectors. As a result, the prevalence of both types of schistosomiasis decreased dramatically from 40% in 1967 to $\leq 3\%$ in 2012. Today, less than 1% of the total population requires preventive chemotherapy. More case studies can be found in Sule et al. 2025 PLOS NTD

The way forward

Prevention of the negative health outcomes caused by schistosomiasis and other water-associated diseases can be achieved by integrating **Health Risk Assessments** into all phases of the **Environmental Impact Assessment** for proposed water management infrastructure. This requires close collaboration with stakeholders, including governmental agencies, developers and contractors, consulting firms, NGOs, and community members.



SOLUTIONS IN THE DESIGN PHASE

To reduce transmission risk, it is crucial to incorporate water, sanitation, and hygiene interventions (**WASH**) in dams' design, including **fish ladders** to allow the passage of migratory predators of snails vectors, including the African river prawn.



CONSTRUCTION PHASE

Establish baseline infection data: Assess baseline snail abundance and distribution, historical and current presence-incidence of schistosomiasis in the human population.

Engage local health and environmental authorities to limit snail proliferation in water bodies and irrigation zones.



OPERATION

Increase surveillance & access to health care: Train physicians and nurses, and provide anthelmintic medication at points of care, plan mass drug administrations, foster education, improve access to clean water, etc.

Environmental control: remove aquatic vegetation, use molluscicides, integrate fish farming into rice agriculture, etc.

Risk of schistosomiasis is **2.4 -2.5x** greater when living close to a dam or an irrigation canal

PHOTO by: Havard Holme (WHO)